

Configuring Chef Server to Monitor Chef Clients

The previous article in DevOps Corner gave an overview of Chef and the procedure to set up a basic sandbox environment with Chef Server. In this article, the authors look at how to install and configure client VMs and Chef agents on them. Subsequently, they move on to explore how Chef agents connect to Chef Server, and how to configure agents to enable monitoring the clients from the server.

So far, we have learnt how to set up Chef Server and configure it by running the `Chef-server-ctl reconfigure` command. We have also learnt how to verify the server configuration by running the test command. Now, the server is almost ready to be used, though there's still some work to be done to get it started and communicating with agents. For this, we need to create an administrator user and organisation. The purpose of creating an administrator user is to control the Chef environment (server and agent), and of creating an organisation is to set up a profile for the Chef environment.

The syntax for a command to create an admin user is as follows:

```
#Chef-server-ctl user-create<username><first name><last name><email><password>--filename<full path of pem file> ---
here pem is the private key file created in the specified folder.
For example:
# Chef-server-ctl user-create magesh magesh kasthuri magesh.kasthuri@wipro.com password --filename /root/.Chef/magesh.pem
```

Then, we can run `org-create` to create the profile for the Chef environment or organisation. The syntax for the command is as follows:

```
# Chef-server-ctl org-create<Org name><Org description>-association-user<Admin user associated with this org>
For example:
# Chef-server-ctl org-create magsandbox "Mag-Sandbox Setup" -association-user magesh
```

After creating the organisation, we can run the test command again to verify if the user is associated properly and the server can start with the organisation profile.

The Chef management console is a Web based tool which provides a graphical user interface (GUI) to manage environments, user roles, cookbooks, etc. After verifying the Chef Server set-up, we need to install the management



console by executing the following commands:

```
# Chef-server-ctl install opscodes-manage
# Chef-server-ctl reconfigure
```

The command to configure and bring up the management console is:

```
# opscodes-manage-ctl reconfigure
```

There are two more steps to complete the server installation and one of them is to install the push job service. This is called the `opscodes-push-jobs-server` and it helps to push jobs or actions to the Chef agent machines. Once this is pushed, the agent will pick them up and run the job or action in the respective machines. This is shown in Figure 2. In this screenshot, you will notice a service called `opscodes-reporting`, which serves reporting data to the management console to enable the user to visualise the operations from the server and agents.

Installing `opscodes-reporting` is the last step in setting up Chef Server and can be done by using the following command:

```
# Chef-server-ctl install opscodes-reporting
```

After installation, run the `reconfigure` command to configure and bring up the service. This will push the reporting data to the management service and enable its display in the Web UI. This is done by the following commands:

```
# Chef-server-ctl reconfigure
# opscodes-reporting-ctl reconfigure
```

For each service, we need to run `Chef-server-ctl reconfigure` as well as the respective `service reconfigure`. This is to enable reconfiguration in both Chef Server and the